

WASP-11b

R-0801

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_221029 · created 2026-07-28T09:14:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459881.8145 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.098 +/- 0.05
Transit depth (Rp/Rs)^2	0.96 +/- 0.99 [%]
Orbital Inclination (inc)	84.86 +/- 1.71
Airmass coefficient 1 (a1)	0.84 +/- 0.56
Airmass coefficient 2 (a2)	0.13 +/- 1.33
Scatter in the residuals of the lightcurve fit is	65.14 %
Best Comparison Star	#2 - [439, 270]
Optimal Aperture	1.46
Optimal Annulus	3.25
Transit Duration (day)	0.032 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.098 ± 0.05 vs 0.131 (deviation 0.51σ)
Duration fit vs published	0.032 d vs 0.1075 d (deviation 1.62σ)
Mid-time O–C	-6.1 min
Depth SNR	1.5
Residual scatter	65.14 %

Light curve

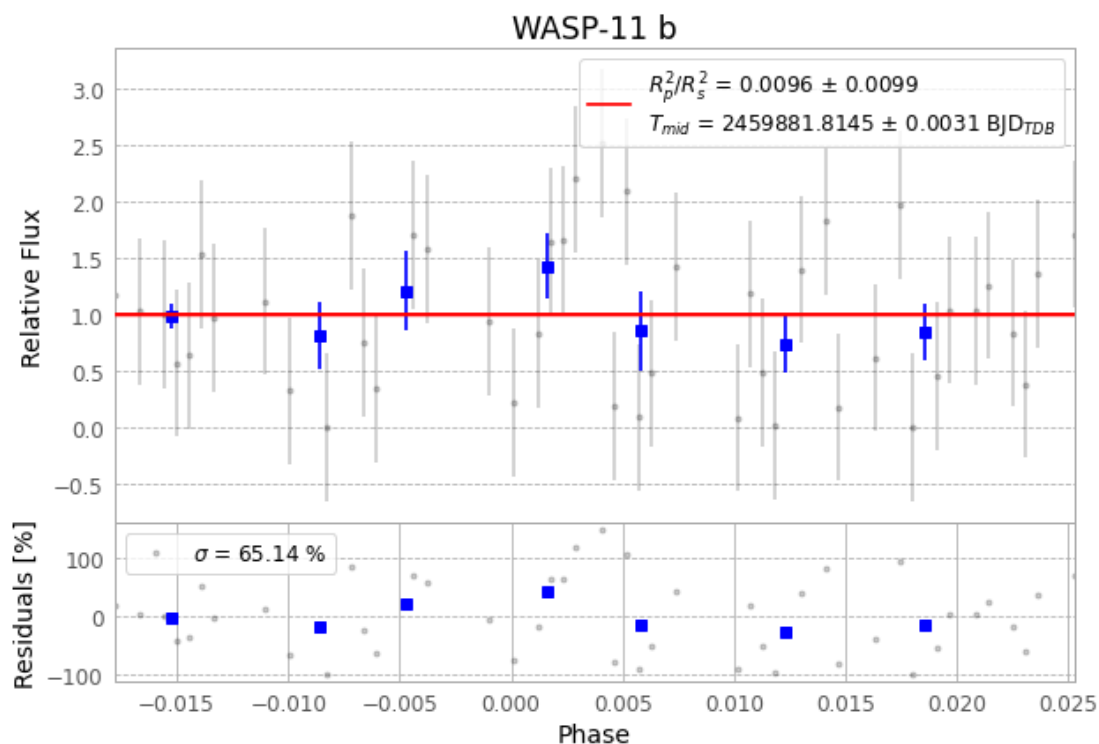


Figure 1 — FinalLightCurve_WASP-11 b_2022-10-28.png (SHA-256 e3b1a42eb0acdfea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [460, 416], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1fbac7fed9b7eb2e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), HATP-10221029054515.FITS → HATP-10221029094816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-221029.log (abc5704d3ebd...), FinalLightCurve_WASP-11 b_2022-10-28.png (e3b1a42eb0ac...), FinalLightCurve_WASP-11 b_2022-10-28.csv (bf5111a1d43d...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 09:14Z.